Interview: Dr. Khanna And The PS3 Gravity Grid

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If you're one of those individuals laboring in the past, refusing to acknowledge and embrace the massive changes in the video game world, we offer you this interview. The world of gaming is now firmly cemented in the world of supercomputing, which is also serving to assist scientists and researchers in ways we never thought possible. By now, you're probably well aware of the **Folding@Home project**, and while that's extremely interesting, this one is about the new frontier...outer space. Is it possible to simulate a collision between two black holes? And if it is, what can we hope to gain from such a project?

Well, here's the **PS3 Gravity Grid**, comprised of a batch of 16 PlayStation 3s and designed to assist in some awfully intriguing research. We wanted to learn more about it, so we had some questions for Dr. Gaurav Khanna, Assistant Professor of Physics at the University of Massachusetts. He was good enough to answer them, and we were awfully impressed with the results. We're not physics experts, but anybody with a vested interest in science - and how the PS3 is contributing to this endeavor - should read on.

PSXE: Could you please explain the overall goal of your project/research? What do you hope to learn by simulating a Black Hole collision (if that is indeed your intent)?

Dr. Khanna: "Black hole collisions are strong sources of 'gravitational waves.' These waves are tiny ripples in the fabric of space-time that travel at the speed of light. They were predicted by Einstein's relativity theory, but have never been directly detected. However, in the last decade or so, there have been several observatories that have been built (for example, NSF's LIGO project in the US and various such installations in Europe & Asia and also the upcoming NASA/ESA LISA Mission) that are supposed to be able to detect these waves.

Detection of these waves, will open a new window into the universe, often referred to as 'gravitational wave astronomy' and will bring us new understanding of some of the most violent events that occur in the universe. My research is on the theory side of the problem - making theoretical predictions of what would be seen by observatories like LIGO, should they witness a binary black hole collision."

Dr. Khanna further points us towards the **official LISA website** and the **LIGO website** as well.

PSXE: What made you choose the PlayStation 3 - i.e., the PlayStation 3 Gravity Grid - for the work? Is it purely for processing power purposes?

Dr. Khanna: "The processing power of the PS3's Cell processor is half of the story. Indeed, the Cell can deliver an incredible amount of computing power, making it very attractive for scientific computation. The other significant factor is the low cost of the Playstation 3 due to various marketing and business reasons. In fact, I am convinced that the PS3 offers a significantly higher 'performance per dollar' compared with anything else that is available in the market today."

PSXE: This isn't the first time the PS3 has been used for science. Are you aware of the Folding@Home project, and if so, do you have any opinions on that?

Dr. Khanna: "Yes, I am well aware of Dr. Pande's Folding@Home public distributed computing project. I think its an excellent use of the PS3's power. I commend Sony and the Folding@Home folks for collaborating and making that happen.

As a quick point to differentiate their approach from ours - all large public distributed computing projects are such that any two tasks are independent i.e. the work being done by one PS3 is quite independent from the work

being done by other. In other words, when two different PS3s are working on their respective tasks, they do not need to communicate with each other to exchange information. This fact makes it possible for the project to progress forward with computational resources distributed all over the globe. However, for other problems (such as my project) it is necessary for each PS3 to exchange information during the computation with the others. This means that each PS3 needs to be able to communicate with the others at a high-speed, therefore it is simply not possible to have them distributed all over the globe.

For this reason, my approach is along the lines of a 'tightly-coupled' traditional Beowulf supercomputing cluster as opposed a 'loosely-coupled' public distributed computing project."

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PSXE: The PS3 is proving to be useful far beyond its gaming capabilities. Do you see more scientists and researchers choosing to use the system in the future?

Dr. Khanna: "Yes, absolutely. In fact, since my cluster caught public attention, I have received queries from several academic and industrial institutions and some are well on their way toward establishing similar clusters."

PSXE: What would you hope your students took away from a project like this, regardless of the outcome? And can we expect to see the world of physics tapping into the world of computing more often in the future?

Dr. Khanna: "Well, the message to the younger generation is that sometimes big opportunities come from very untraditional directions, and one should always stay alert and aware of what is going on in other areas. We have a very fast changing world and one has to be willing to learn new

things throughout and be creative.

Computing is making a stronger and stronger impact on the sciences. Sometimes it is preferable to perform a 'simulated experiment' even when it is possible to do something similar in the laboratory, simply because of the steep cost associated to the laboratory research. Of course, in my area, simulation plays an even larger role - we simply can't take two black holes and have them collide in a laboratory!"

PSXE: Would the work you're doing now have been possible before the PS3? In other words, did this project require such a technological breakthrough?

Dr. Khanna: "So far, I had been doing my work using national supercomputing centers (SDSC, TACC, etc.). However, these are shared facilities, with long wait times, and require various research proposals, reports and approvals. With my PS3 cluster, I can now do most of my work right here in my laboratory. Plus, the cluster is dedicated to my research and is therefore always readily available.

The technological breakthrough is of a somewhat different nature here: With a very small investment, anyone can have access to supercomputing level performance for whatever research problem one is attempting to tackle. This is the model that has proved to be a success by what we have been able to do with our PS3 Gravity Grid."

PSXE: We are a gaming site, so we just have to ask- are you or anybody else involved with this work a gamer? And if so, what types of games do you enjoy playing?

Dr. Khanna: "I certainly do boot into the PS3's game partition now and then! I really do enjoy Tekken and also many of the car racing games (Paradise City [aka Burnout Paradise], MotorStorm, GT, etc). I not very good at them though :-(."

We certainly appreciate Dr. Khanna taking the time to answer our questions, and we really hope to see more projects like this in the future. While we love our games, we think it's just amazing that video game consoles are capable of such scientific feats. And always remember, this universe is a very *big* place, and there's a lot more to it than ring-tones and your girlfriend's birthday. ...well, don't forget the birthday, but keep that mind open!

For more information on Dr. Khanna's project, be sure to visit the **PS3 Gravity Grid website**. It's well worth your time.